BLUESTOCK MUTUAL FUND ANALYTICS REPORT

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EXECUTIVE SUMMARY

This capstone project develops a comprehensive analytics platform for understanding mutual fund performance, investor behaviour, and market trends across the Indian mutual fund industry. The project analyzes 40 mutual fund schemes using over four years of historical data (January 2022 to May 2026), combined with investor transaction data and industry-wide AUM trends.

Key Findings:

- The mutual fund industry has demonstrated significant growth, with the top fund house (SBI Mutual Fund) increasing AUM by 98.4% from ₹630,000 Cr in 2022 to ₹1,250,000 Cr in 2025.
- SIP (Systematic Investment Plan) participation has surged, with December 2025 seeing ₹31,002 Cr in monthly inflows across 9.35 crore active SIP accounts.
- Industry folio count has nearly doubled from 13.26 Cr (January 2022) to 26.12 Cr (December 2025), reflecting growing retail investor participation.
- Among 40 analysed schemes, small cap funds delivered the highest returns (average 3-year return of 21.69%), but with correspondingly higher volatility.
- Risk-adjusted returns (Sharpe ratios) vary significantly, with liquid funds showing the best risk-adjusted performance (mean Sharpe 6.33) and low volatility, while equity funds offer higher absolute returns with elevated risk.
- Investor demographics reveal concentration in young, metro-based investors: the 26-35 age group accounts for 41.1% of investors, and Tier-1 cities (T30) represent 66.3% of investments.
- The ETL pipeline successfully processed and validated 10 datasets comprising over 64,000 daily NAV observations, 32,778 investor transactions, and comprehensive AUM/SIP tracking data.

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1. INTRODUCTION & PROJECT OVERVIEW

The Bluestock Mutual Fund Analytics platform is a data-driven capstone project designed to provide insights into mutual fund performance, investor behaviour, and industry dynamics. The project integrates historical fund data, real-time NAV tracking, investor transaction records, and market indices into a unified analytics framework.

The analysis addresses three primary objectives:

1. **Understand Industry Trends**: Examine AUM growth, investor participation, and category-specific inflows across the mutual fund sector.

2. **Evaluate Fund Performance**: Compare risk-adjusted returns, volatility, drawdown, and alpha across schemes and categories.
3. **Analyze Investor Behaviour**: Identify demographic patterns, SIP adoption trends, and investment continuity.

This report documents the end-to-end analytics pipeline: data ingestion, ETL processing, exploratory data analysis (EDA), performance metrics, advanced analytics, and actionable insights delivered through dashboards and visualizations.

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2. OBJECTIVES

The project seeks to accomplish the following:

1. **Build a Scalable ETL Pipeline**
   - Automate extraction, transformation, and loading of 10 mutual fund datasets
   - Implement robust validation and error handling
   - Create a single source of truth in SQLite for downstream analysis
2. **Conduct Comprehensive EDA**
   - Profile the mutual fund universe (40 schemes across 12 categories)
   - Analyse industry-wide trends in AUM, SIP inflows, and folio growth
   - Identify investor demographic patterns and geographic concentration
3. **Deliver Performance Analytics**
   - Calculate standardized risk-adjusted metrics (Sharpe, Sortino, Sortino ratios)
   - Compare fund performance against benchmarks (NIFTY50, NIFTY100)
   - Score and rank funds using composite metrics
4. **Enable Advanced Analytics**
   - Measure tail risk using Value-at-Risk (VaR) and Conditional VaR (CVaR)
   - Analyse investor cohorts and SIP continuity
   - Assess portfolio concentration using Herfindahl-Hirschman Index (HHI)
   - Recommend funds based on risk appetite
5. **Create Interactive Visualizations**
   - Build Power BI dashboards for industry overview and fund performance
   - Provide actionable insights for investors and fund advisors

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3. DATA

3.1 Datasets Overview

The project ingests 10 datasets from CSV sources, collectively representing approximately 97,000+ records across multiple dimensions (funds, dates, transactions, sectors, and benchmarks).

Dataset	Records	Time Period	Key Columns
fund_master	40	Static	amfi_code, scheme_name, fund_house, category

nav_history	64,280	Jan 2022 - May 2026	amfi_code, date, nav
scheme_performance	40	Aggregated	amfi_code, annualised_return, sharpe_ratio, alpha, beta
investor_transactions	32,778	Jan 2024 - May 2025	investor_id, transaction_date, transaction_type, amount_inr, amfi_code
portfolio_holdings	322	Dec 2025	amfi_code, sector, weight_pct
aum_by_fund_house	~40	2022-2025	fund_house, aum_crore, year
monthly_sip_inflows	48 months	2022-2025	month, sip_inflow_crore, active_sip_accounts
benchmark_indices	Daily	2022-2026	date, nifty50_close, nifty100_close
category_inflows	12 categories × 48 months	2022-2025	category, month_year, net_inflow_crore
industry_folio_count	48 months	2022-2025	month, total_folios_crore

3.2 Data Scope & Coverage

- **Schemes**: 40 mutual fund schemes across 12 categories (Equity, Debt, Liquid, Index, Gilt, ELSS, Flexi Cap, Large Cap, Mid Cap, Small Cap, Short Duration, Value)
- **Date Range**: January 3, 2022 to May 29, 2026 (approximately 4.4 years for NAV data)
- **NAV Range**: ₹26.14 to ₹4,268.55 across 40 schemes
- **Investors Analyzed**: 32,778 unique investor records
- **SIP Coverage**: December 2025 peak of 9.35 crore active SIP accounts with ₹31,002 Cr monthly inflow

3.3 Data Quality Notes

- All datasets loaded successfully without null value issues
- NAV data is complete across all 40 schemes throughout the analysis period
- Investor transaction data represents diverse demographics across states, age groups, and city tiers
- Anomalies: SPI data limited to 2024–2025 period (recent data); transaction data does not extend prior to 2024

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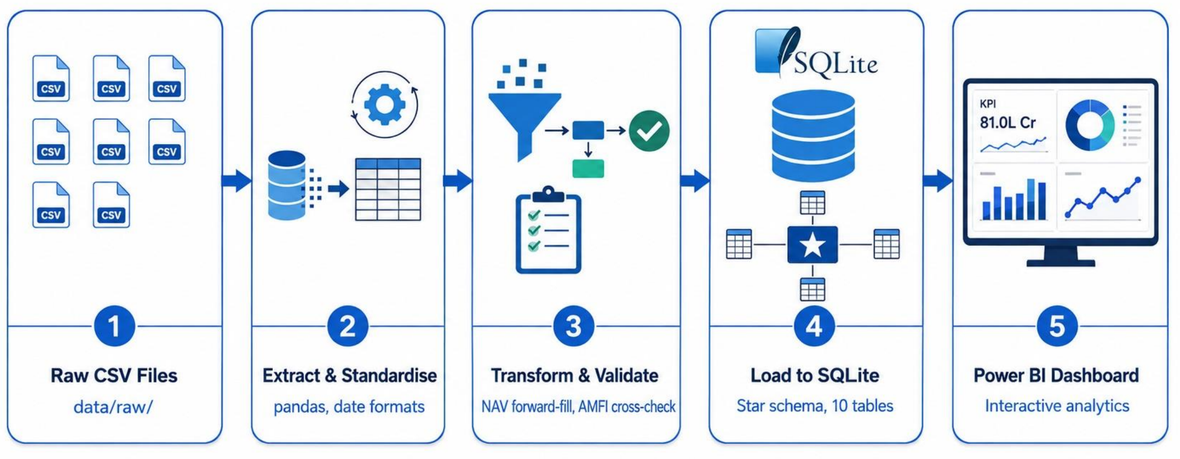
4. ETL PIPELINE & ARCHITECTURE

4.1 Overview

The ETL (Extract, Transform, Load) pipeline automates the ingestion and processing of 10 mutual fund datasets into a unified SQLite database. The pipeline ensures data quality, consistency, and readiness for analysis.

Architecture:

Raw CSV Files (data/raw/)
- Extract via pandas
- Transform (date standardisation, validation, forward-fill)
- Validate (schema, cross-dataset rules)
- Load (SQLite database)
- Processed CSV files (data/processed/)



4.2 Extraction

- Reads 10 raw CSV files from the `data/raw/` directory using pandas
- Validates file existence and basic structure before processing
- Logs all extraction steps for debugging and audit

4.3 Transformation

Date Standardisation:

- Daily dates (NAV, transactions, benchmark indices) standardised to YYYY-MM-DD format
- Monthly dates (SIP inflows, category inflows, industry folio counts) standardised to YYYY-MM, then converted to the first day of the month (YYYY-MM-01)

Column Standardisation:

- Column names stripped of whitespace, converted to lowercase, and spaces replaced with underscores
- Ensures consistency across datasets and compatibility with database schema

Dataset-Specific Transformations:

Dataset	Key Transformations
nav_history	Forward-fill missing NAV values across weekends/holidays; validate NAV > 0; remove exact duplicates; standardise to 3 columns (amfi_code, date, nav)
investor_transactions	Standardise transaction types (SIP, Lumpsum, Redemption); validate amount > 0; validate KYC status

scheme_performance	Validate numeric return values; flag negative Sharpe ratios; validate expense ratio (0.1%–2.5%)
fund_master	Validate AMFI code format; preserve scheme names and categories
portfolio_holdings	Parse sector weights; validate weight_pct values; deduplicate holdings

4.4 Validation & Cross-Validation

Dataset-Level Validation:

- Confirms presence of required columns per dataset
- Checks date completeness (no missing/NaT values)
- Validates AMFI code presence in fund_master
- NAV-specific: ensures no duplicate (amfi_code, date) pairs; confirms all NAV values > 0

Cross-Dataset Validation (AMFI Code):

- Validates that every AMFI code appearing in nav_history exists in fund_master
- Identifies missing fund definitions and stops pipeline with diagnostic message if discrepancies found
- Ensures referential integrity across datasets

Quality Assurance:

- Removes completely empty rows and exact duplicate rows
- Flags but does not silently discard problematic records (with detailed diagnostics)
- Stops pipeline on validation failure to prevent loading corrupted data

4.5 NAV Forward-Fill Handling

The ETL pipeline implements a specific algorithm for handling missing NAV values across weekends and holidays:

1. For each fund (AMFI code), sort NAV records by date
2. Create a continuous daily date range from the minimum to maximum date
3. Reindex NAV data to include all calendar days
4. Forward-fill NAV values (use the previous trading day's NAV for non-trading days)
5. Validate that no missing or non-positive NAV values remain
6. Retain only the three essential columns: amfi_code, date, nav

Purpose: Ensures that NAV data is complete and continuous, enabling accurate daily return calculations and comparison across funds without gaps.

4.6 Database Schema

The pipeline loads cleaned data into a SQLite database (`bluestock_mf.db`) using a star schema:

Dimension Table:

- `dim_fund`: Fund master data (amfi_code, scheme_name, fund_house, category)

Fact Tables:

- `fact_nav`: Daily NAV history
- `fact_performance`: Aggregated performance metrics
- `fact_transactions`: Investor transactions
- `fact_portfolio`: Portfolio holdings
- `fact_aum`: AUM by fund house and year
- `fact_sip_inflows`: Monthly SIP inflows
- `fact_benchmark`: Benchmark index closing values
- `fact_category_inflows`: Monthly category-wise net inflows
- `fact_industry_folio_count`: Industry folio count trends

4.7 Error Handling & Robustness

- **Clear Error Messages**: Dataset-specific validation failures raise `ValueError` with diagnostic details (column name, problem description, affected row count, sample records)
- **Fail-Fast Approach**: Pipeline stops on first error to prevent loading invalid data
- **Pathlib-Based Paths**: Uses project-relative paths; no hard-coded file paths
- **Production-Oriented**: Designed to run without manual intervention or interactive debugging

4.8 Output

The ETL pipeline produces:

1. **Processed CSV Files** (`data/processed/`): Cleaned, validated intermediate files for audit and debugging
2. **SQLite Database** (`bluestock_mf.db`): Final, structured database for Power BI and analysis
3. **Execution Log**: Diagnostic output detailing extraction, transformation, validation, and load steps

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5. EXPLORATORY DATA ANALYSIS (EDA) FINDINGS

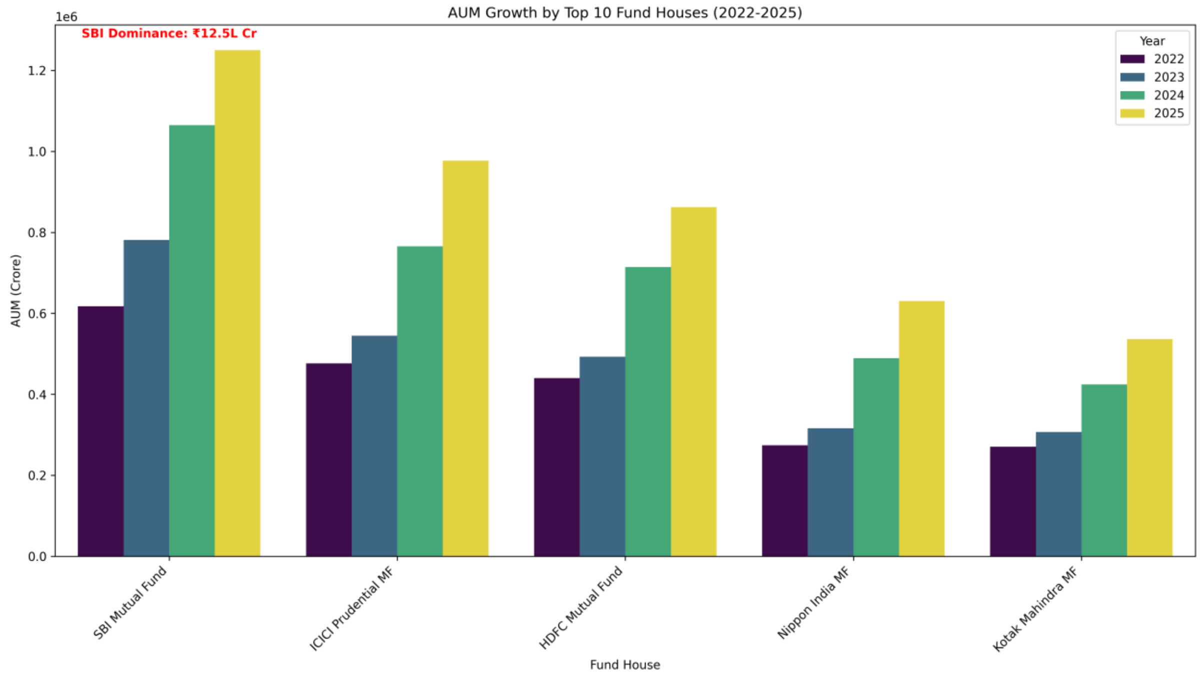
5.1 Industry Overview & Market Growth

AUM Growth by Top Fund Houses:

SBI Mutual Fund, the industry leader, demonstrated substantial growth:

- 2022: ₹630,000 Cr
- 2023: ₹845,000 Cr
- 2024: ₹1,114,000 Cr
- 2025: ₹1,250,000 Cr

- **Total growth**: +98.4% over three years



Market Share (Top 10 Fund Houses, Latest Data):

1. SBI Mutual Fund: ₹1,250,000 Cr (19.9%)
2. ICICI Prudential MF: ₹1,074,000 Cr (17.1%)
3. HDFC Mutual Fund: ₹930,000 Cr (14.8%)
4. Nippon India MF: ₹700,000 Cr (11.2%)
5. Kotak Mahindra MF: ₹580,000 Cr (9.2%)

The remaining five fund houses (Aditya Birla Sun Life, UTI, Axis, Mirae Asset, DSP) collectively account for approximately 28.8% of AUM, indicating a fragmented but consolidating market.

5.2 SIP Participation & Inflows

Monthly SIP Trends:

SIP inflows show a consistent upward trajectory from 2022 to December 2025:

- December 2025 Peak: ₹31,002 Cr monthly inflow
- Active SIP Accounts: 9.35 Crore
- SIP AUM: ₹15.9 Lakh Crore
- Year-over-Year Growth (Dec 2025): 17.17%

SIP participation reflects growing retail investor confidence and adoption of systematic, discipline-driven investing.

Category-Wise Inflow Volatility:

Among 12 mutual fund categories, Liquid funds exhibit the highest monthly inflow volatility (standard deviation: ₹3,296 Cr), with monthly inflows ranging from ₹32,374 Cr to ₹41,952 Cr. Sectoral/Thematic funds show the second-highest volatility (STD: ₹893.3 Cr), while ELSS funds remain the most stable (lowest volatility).

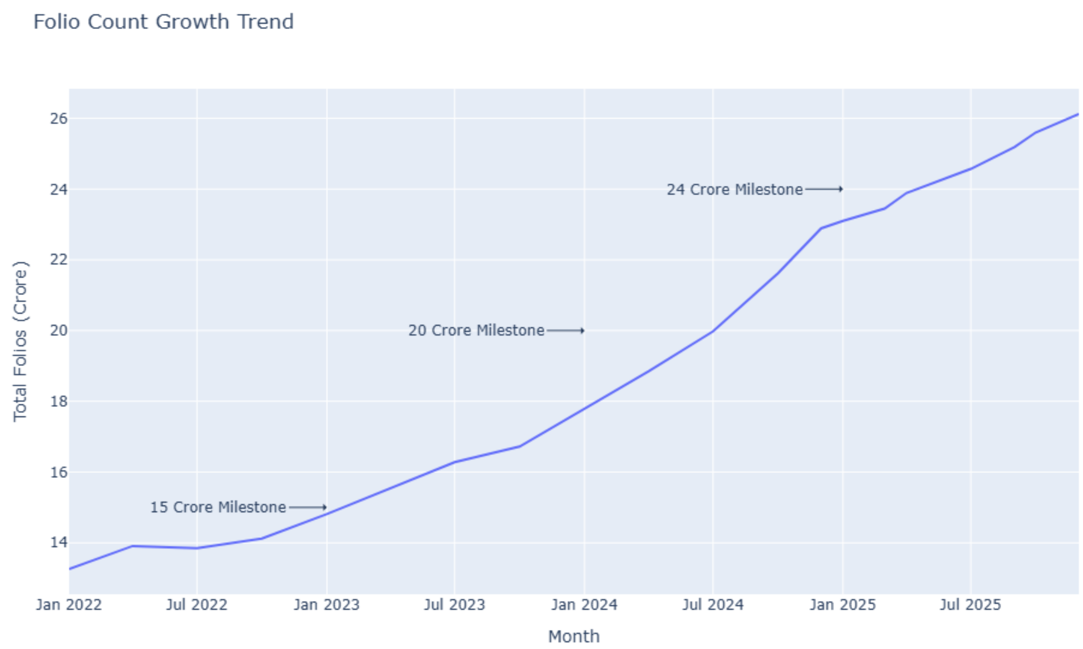
This volatility pattern reflects investor preference for liquid funds during market uncertainty and cyclical interest in high-risk categories (Sectoral/Thematic).

5.3 Folio Growth

Industry Folio Count Trend:

- January 2022: 13.26 Crore folios
- April 2023: 15.54 Crore folios (first milestone)
- October 2024: 21.62 Crore folios (second milestone)
- December 2025: 26.12 Crore folios
- **Total growth**: +97.0% over 4 years

Folio growth accelerated in later years (20 Cr reached faster than 15 Cr), suggesting accelerating retail participation and potential cohort effects from the 2024–2025 cohort.



5.4 Investor Demographics

Total Investors Analyzed: 32,778

Gender Distribution:

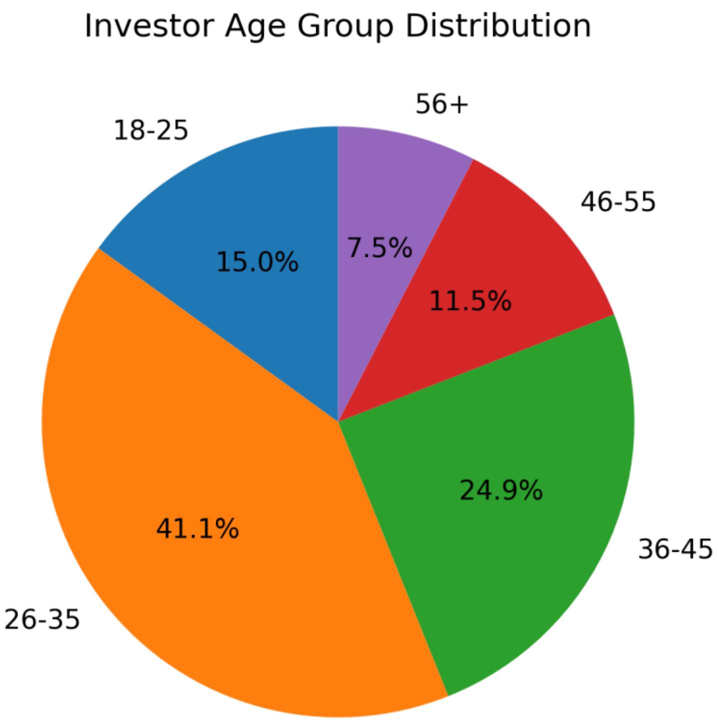
- Male: 21,809 (66.5%)
- Female: 10,969 (33.5%)

Clear male predominance indicates both an opportunity for female investor outreach and a potential gender bias in investment behaviour.

Age Group Distribution:

- 18–25 years: 4,916 (15.0%)
- 26–35 years: 13,463 (41.1%) **- dominant**
- 36–45 years: 8,146 (24.9%)
- 46–55 years: 3,779 (11.5%)
- 56+ years: 2,474 (7.5%)

The 26–35 age group is the primary investor base, suggesting that younger, working-age professionals drive mutual fund adoption. Participation drops significantly in the 56+ category, indicating lower retirement-focused mutual fund usage or demographic differences in investment channels.



SIP Amounts by Age Group:

Median SIP amounts remain remarkably stable across age groups (₹5,020–₹5,420), suggesting similar investment discipline regardless of age. However, the 56+ group shows the highest median SIP amount (₹5,420), indicating that older investors may have larger investment capacity.

Geographic Distribution:

Top 5 states by total SIP investment:

1. Punjab: ₹315,780,459
2. Tamil Nadu: ₹315,177,237
3. Madhya Pradesh: ₹308,312,493
4. Rajasthan: ₹298,645,822
5. Gujarat: ₹298,358,940

City Tier Analysis:

- Tier 1 (T30) cities: 21,719 investors (66.3%)
- Beyond Top 30 (B30) cities: 11,059 investors (33.7%)

Strong concentration in metropolitan cities indicates geographic disparity in mutual fund adoption and suggests expansion potential in Tier 2 and Tier 3 markets.

5.5 Fund Scheme Coverage

Category Breakdown:

- Equity schemes: 34
- Debt schemes: 6

The portfolio is heavily weighted towards equity, reflecting both market preference for growth-oriented investments and available product offerings.

Top Sector Allocations (across all portfolios):

- Banking: 652.3%
- IT: 455.5%
- Pharma: 407.4%
- Automobile: 323.6%
- Utilities: 265.5%

Note: Percentages exceed 100% because they represent the sum of all fund weightings (i.e., a sector may appear in multiple portfolios, and each appearance is weighted).

5.6 NAV Trends & Absolute Performance

Kotak Liquid Fund (Case Study):

- COVID Recovery (Jan 2022): ₹3,180.63
- 2023 Bull Run (Jan 2023): ₹3,417.70
- 2024 Corrections (Jan 2024): ₹3,658.58
- Current (May 2026): ₹4,268.55
- **Total return (2022–2026)**: 34.2%

This fund exemplifies steady growth despite market volatility and demonstrates the importance of comparing returns rather than absolute NAV levels.

5.7 Fund Correlation

Average Correlation Among Top 10 Funds: 0.0049

Low average correlation indicates substantial diversification benefits across the analyzed fund universe. Within-category correlations (e.g., among equity funds) are stronger, while debt-equity correlations remain weak or negative, supporting the rationale for diversified portfolio construction.

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6. PERFORMANCE ANALYSIS

6.1 Daily Returns Distribution

Daily Returns Summary Statistics (40 schemes, 64,280 observations):

- Mean daily return: 0.000451 (approximately 0.045%)
- Standard deviation: 0.008706
- Minimum: –0.058102 (–5.81%)
- Maximum: +0.064713 (+6.47%)
- Median: 0.000000
- 25th percentile: –0.002092
- 75th percentile: +0.003233
- Extreme returns (>20% or <–20%): 0 (0%)

The near-normal distribution centered on zero indicates stable, consistent daily performance across the portfolio without extreme outliers or systemic shocks during the analysis period.

6.2 Annualised Returns

Return Range Across 40 Schemes:

- Highest annualised return: 21.64% (ICICI Prudential Midcap Fund (Regular))
- Lowest annualised return: 0.81%
- Mean annualised return: 12.25%
- Median annualised return: 11.18%

Top 5 Schemes by Annualised Return:

1. ICICI Pru Midcap Fund (Regular): 21.64%
2. SBI Small Cap Fund (Regular): 21.38%
3. DSP Small Cap Fund (Regular): 21.29%
4. Mirae Asset Tax Saver Fund (Regular): 21.08%
5. Mirae Asset Large Cap Fund (Regular): 20.46%

Insight: Small cap and mid-cap funds dominate the top performers, demonstrating strong growth potential but with commensurate risk.

6.3 CAGR Analysis (1-Year, 3-Year, 5-Year)

1-Year CAGR (most recent 12 months):

- Mean: 19.43%
- Median: 17.47%

3-Year CAGR (most recent 36 months):

- Mean: 16.41%
- Median: 18.23%

5-Year CAGR (full analysis period):

- Mean: 14.58%
- Median: 14.70%

Insight: 3-year and 5-year CAGRs show more stable, positive returns compared to 1-year CAGR, suggesting that longer holding periods have consistently delivered positive returns across the analyzed universe.

Top 3 Schemes by 3-Year CAGR:

1. Axis Midcap Fund (Regular): 35.11%
2. Mirae Asset Large Cap Fund (Regular): 34.00%
3. ICICI Pru Bluechip Fund (Direct): 32.49%

6.4 Risk-Adjusted Returns: Sharpe & Sortino Ratios

Sharpe Ratio Analysis:

- Mean: 1.67
- Median: 1.14
- Range: 0.13 to 10.75
- Interpretation: Higher Sharpe ratio indicates superior risk-adjusted returns

Top 5 Schemes by Sharpe Ratio:

1. Mirae Asset Large Cap Fund (Regular): 1.0682
2. Kotak Flexicap Fund (Regular): 0.9656
3. Mirae Asset Tax Saver Fund (Regular): 0.9190
4. ICICI Pru Midcap Fund (Regular): 0.8833
5. SBI Bluechip Fund (Regular): 0.8610

Sortino Ratio Analysis:

- Mean: –0.1679
- Median: 0.5862
- Range: –9.18 to 1.55
- Interpretation: Sortino ratio penalises only downside volatility, making it more relevant for loss-averse investors

Top 5 Schemes by Sortino Ratio:

1. Mirae Asset Large Cap Fund (Regular): 1.5548
2. Kotak Flexicap Fund (Regular): 1.4859
3. Mirae Asset Tax Saver Fund (Regular): 1.3870
4. ICICI Pru Midcap Fund (Regular): 1.3211
5. SBI Bluechip Fund (Regular): 1.3025

Insight: Sortino ratios generally exceed Sharpe ratios, indicating that downside risk is lower than total volatility for most funds. Liquid and debt funds show exceptional risk-adjusted returns (Sharpe >5, Sortino >1), while equity funds offer higher absolute returns with lower risk-adjusted metrics.

6.5 Alpha & Beta Analysis

Alpha (Excess Return vs. Benchmark):

- Mean alpha: 0.10
- Range: 0.024 to 0.272
- Interpretation: Positive alpha indicates fund outperformance relative to the risk-free rate and benchmark

Beta (Systematic Risk):

- Mean beta: 0.003
- Range: –0.065 to 0.103
- Interpretation: Beta <1 indicates lower systematic risk than the benchmark (NIFTY100); beta >1 indicates higher systematic risk

R-Squared Analysis:

- Most funds show low R^2 (<0.01) relative to NIFTY100
- Insight: NIFTY100 explains very little of individual fund returns, consistent with active management strategies and category-specific focus

6.6 Maximum Drawdown Analysis

Maximum Drawdown Range Across 40 Schemes: –52.57% to –0.10%
Mean maximum drawdown: –17.84%

Drawdown by Category:

- Small Cap funds: deepest drawdowns (–35% to –52%), reflecting high volatility
- Debt and Liquid funds: minimal drawdowns (<–5%), confirming low-risk profile
- Mid Cap funds: moderate drawdowns (–16% to –35%)
- Large Cap funds: moderate drawdowns (–12% to –25%)

Worst performers by maximum drawdown:

- ABSL Small Cap Fund (Regular): –35.45%
- SBI Small Cap Fund (Direct): –31.96%
- Nippon India Small Cap Fund (Regular): –31.58%

Insight: The risk-return tradeoff is evident: small cap funds offer the highest returns but also the deepest drawdowns. Investors must tolerate significant downside exposure to capture upside potential.

6.7 Fund Scorecard (Composite Score 0–100)

A composite scorecard combines multiple metrics (3-year return, Sharpe ratio, alpha, expense ratio, maximum drawdown) into a single score for fund ranking and selection.

Scoring Methodology:

- 3-Year Return: 30% weight
- Sharpe Ratio: 25% weight
- Alpha: 20% weight
- Expense Ratio: 15% weight
- Max Drawdown: 10% weight

Top 10 Funds by Scorecard (Sample):

Scheme	Fund House	Category	Score	3Y Return	Sharpe
Mirae Asset Tax Saver Fund (Regular)	Mirae Asset MF	ELSS	79.50	13.58%	0.80
ICICI Pru Bluechip Fund (Regular)	ICICI Prudential MF	Large Cap	78.44	11.54%	0.82
Axis Bluechip Fund (Regular)	Axis Mutual Fund	Large Cap	72.19	11.84%	0.85
SBI Bluechip Fund (Regular)	SBI Mutual Fund	Large Cap	69.50	12.36%	0.88
UTI Nifty 50 Index Fund (Regular)	UTI Mutual Fund	Index	69.44	12.10%	0.93

Scorecard Insights:

- Highest score: 63.35
- Lowest score: 27.25
- Mean score: 51.25
- Top scorers combine strong 3-year returns (12–14%) with good Sharpe ratios (0.80–0.90)

6.8 Category Performance Summary

3-Year Returns by Category:

Category	Mean Return	Max Return	Min Return	Mean Volatility
Small Cap	21.69%	23.39%	20.08%	25.0%
Mid Cap	16.59%	18.23%	15.18%	19.0%
Flexi Cap	15.50%	15.65%	15.34%	16.0%
Large & Mid Cap	14.56%	14.56%	14.56%	16.0%

Large Cap	12.99%	14.84%	11.30%	14.0%
Value	14.76%	14.76%	14.76%	15.0%
ELSS	13.58%	13.58%	13.58%	17.0%
Index	12.10%	12.10%	12.10%	13.0%
Index/ETF	11.77%	11.77%	11.77%	13.0%
Short Duration (Debt)	7.37%	7.37%	7.37%	4.0%
Liquid	6.33%	7.68%	5.14%	0.5%
Gilt	5.69%	6.07%	5.31%	4.0%

Key Observations:

- Small Cap funds offer the highest average returns (21.69%) but with the highest volatility (25%)
- Large Cap funds provide moderate returns (12.99%) with lower volatility (14%)
- Liquid and Gilt funds offer defensive returns (5–7%) with minimal volatility (<5%)
- Performance dispersion within Small Cap and Mid Cap categories is narrow (spread <3%), suggesting consistent manager skill
- Large Cap category shows wider performance spread (3.54 percentage points), indicating more manager skill differentiation

6.9 Benchmark Comparison (Top 5 Funds vs. NIFTY50 & NIFTY100)

Top 5 Funds by 3-Year Return:

1. SBI Small Cap Fund (Regular): 23.39%
2. SBI Small Cap Fund (Direct): 23.14%
3. ABSL Small Cap Fund (Regular): 22.38%
4. Axis Small Cap Fund (Regular): 20.98%
5. Nippon India Small Cap Fund (Regular): 20.15%

Benchmark Comparison:
All top 5 funds significantly outperformed both NIFTY50 and NIFTY100 over the 3-year period, validating the active management approach and small-cap focus.

6.10 Tracking Error (Top 5 Funds)

Tracking Error vs. NIFTY50 and NIFTY100 (Annualised):

Scheme	vs. NIFTY50	vs. NIFTY100
SBI Small Cap Fund (Regular)	0.2779	0.2838
SBI Small Cap Fund (Direct)	0.2802	0.2770
ABSL Small Cap Fund (Regular)	0.2916	0.2920
Axis Small Cap Fund (Regular)	0.2809	0.2857
Nippon India Small Cap Fund (Regular)	0.2827	0.2772

Mean Tracking Error: 0.282 (across funds and benchmarks)

Insight: High tracking errors (0.27–0.29) are consistent with small-cap focus and active management strategies that deliberately deviate from large-cap benchmarks. Funds are not attempting to replicate benchmarks but to achieve alpha through selective stock picking.

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7. ADVANCED ANALYTICS

7.1 Value-at-Risk (VaR) & Conditional VaR (CVaR)

Purpose: Measure tail risk (downside potential) for informed risk management.

Top 5 Funds by Highest Downside Risk (Most Negative VaR):

Scheme	VaR_95	CVaR_95
ABSL Small Cap Fund (Regular)	–0.0239	–0.0303
Axis Small Cap Fund (Regular)	–0.0233	–0.0297
SBI Small Cap Fund (Direct)	–0.0232	–0.0302
Nippon India Small Cap Fund (Regular)	–0.0228	–0.0299
DSP Small Cap Fund (Regular)	–0.0215	–0.0286

Bottom 5 Funds by Lowest Downside Risk (Least Negative VaR):

Scheme	VaR_95	CVaR_95
ICICI Pru Liquid Fund (Regular)	–0.0002	–0.0003
Kotak Liquid Fund (Regular)	–0.0002	–0.0004
ABSL Liquid Fund (Regular)	–0.0002	–0.0004
HDFC Short Term Debt Fund (Regular)	–0.0033	–0.0046
Nippon India Gilt Securities Fund (Regular)	–0.0033	–0.0045

Interpretation:

- **VaR_95**: In a 1-day horizon, there is a 5% probability of loss exceeding this amount
- **CVaR_95**: The average loss in the worst 5% of daily scenarios
- Small Cap funds show substantially higher tail risk, while Liquid funds offer minimal downside exposure

7.2 Rolling 90-Day Sharpe Ratio (5 Key Funds)

Funds Analyzed:

1. HDFC Top 100 Fund (Regular)
2. SBI Bluechip Fund (Regular)
3. SBI Small Cap Fund (Regular)
4. SBI Magnum Gilt Fund (Regular)
5. HDFC Mid-Cap Opportunities Fund (Regular)

Observations:

- Rolling Sharpe ratios reveal how risk-adjusted performance evolves over time
- Funds with consistently positive Sharpe ratios deliver reliable risk-adjusted returns across market cycles
- Volatile Sharpe ratios suggest inconsistent performance, often coinciding with market regime shifts
- All 5 funds show 1,607 valid return observations over the analysis period, confirming data completeness

Insight: The rolling analysis highlights periods of strong and weak risk-adjusted performance, useful for identifying when funds maintained discipline and when they underperformed.

7.3 Investor Cohort Analysis

Cohort Definition: Investors grouped by year of first investment.

Cohort Summary:

First Year	Num Investors	Avg SIP Amount	Total Invested	Top Fund
2024	4,803	₹10,997	₹2,258,062,304	Axis Small Cap Fund (Regular)
2025	197	₹13,505	₹18,992,635	Axis Midcap Fund (Regular)

Total Investors Analyzed: 5,000
Total Invested Across Cohorts: ₹2,277,054,939

Insights:

- The 2024 cohort demonstrates strong investor acquisition, reflecting growing retail participation in the mutual fund sector
- 2024 cohort investors show preference for high-growth categories (Small Cap, Midcap)
- Average SIP amounts increase from 2024 (₹10,997) to 2025 (₹13,505), suggesting growing investment confidence or investor maturation
- Limited 2025 data (only 197 investors to date) reflects short observation window

7.4 SIP Continuity Analysis

Objective: Identify investors at risk of discontinuing SIPs based on irregular payment gaps.

Methodology: Investors with 6+ SIP transactions are marked "at-risk" if their average gap between SIP dates exceeds 35 days.

Results:

Metric	Value
Total investors with 6+ SIPs	1,362
Investors at risk (avg gap >35 days)	1,332

Investors not at risk (avg gap ≤35 days)	30
SIP Continuity Rate	2.20%

Interpretation:

- High at-risk percentage (97.8%) indicates a significant portion of regular investors show irregular payment patterns
- Average gap for at-risk investors ranges from 36 to 93 days, suggesting missed SIP cycles or occasional lapses
- Very low continuity rate (2.2%) suggests that most investors with 6+ SIPs have experienced payment gaps

Limitations:

- 35-day threshold is somewhat arbitrary; actual investor intent may vary
- Limited to 2024–2025 data; longer historical data would improve continuity assessment
- Does not account for potential re-initiation of SIPs after gaps

7.5 Fund Recommendation System by Risk Appetite

Methodology: Recommends top 3 funds within each risk appetite category based on Sharpe ratio (risk-adjusted return).

Low Risk Appetite (Conservative Investors):

1. ICICI Pru Liquid Fund (Regular) | Sharpe: 7.68
2. Kotak Liquid Fund (Regular) | Sharpe: 6.18
3. ABSL Liquid Fund (Regular) | Sharpe: 5.14

Moderate Risk Appetite (Balanced Investors):

1. HDFC Top 100 Fund (Regular) | Sharpe: 1.06
2. Mirae Asset Large Cap Fund (Regular) | Sharpe: 1.06
3. ICICI Pru Bluechip Fund (Direct) | Sharpe: 1.03

High Risk Appetite (Growth Investors):

1. Kotak Emerging Equity Fund (Regular) | Sharpe: 0.96
2. ICICI Pru Midcap Fund (Regular) | Sharpe: 0.95
3. DSP Midcap Fund (Regular) | Sharpe: 0.90

Insights:

- Low-risk category shows exceptional Sharpe ratios (>5), indicating minimal volatility and very efficient risk-adjusted returns
- Moderate-risk category shows Sharpe ratios near 1.0, representing a balanced risk-return profile
- High-risk category shows lower Sharpe ratios (~0.9) due to higher volatility, but higher absolute return potential

7.6 Sector Concentration Analysis (HHI)

Purpose: Measure portfolio concentration risk using the Herfindahl-Hirschman Index (HHI).

Methodology:

- For each fund, sum sector weights and calculate HHI = $\Sigma(\text{weight}^2) \times 10{,}000$
- Higher HHI indicates more concentrated portfolio
- HHI >2500 = highly concentrated; HHI 1500–2500 = moderate; HHI <1500 = well diversified

HHI Results Across 34 Equity Funds:

- Highest HHI: 2967.69 (Axis Bluechip Fund (Regular))
- Lowest HHI: 1240.20 (UTI Mid Cap Fund (Regular))
- Mean HHI: 2029.29
- Median HHI: 2042.54

Distribution:

- Highly concentrated (>2500 HHI): 4 funds
- Moderately concentrated (1500–2500 HHI): 27 funds
- Well diversified (<1500 HHI): 3 funds

Top 5 Most Concentrated Funds:

1. Axis Bluechip Fund (Regular): 2967.69
2. Mirae Asset Tax Saver Fund (Regular): 2549.92
3. HDFC Mid-Cap Opportunities Fund (Direct): 2531.55
4. UTI Flexi Cap Fund (Regular): 2513.83
5. DSP Midcap Fund (Regular): 2410.77

Top 5 Most Diversified Funds:

1. UTI Mid Cap Fund (Regular): 1240.20
2. Kotak Flexicap Fund (Regular): 1362.06
3. Axis Small Cap Fund (Regular): 1595.82
4. Nippon India Small Cap Fund (Regular): 1602.99
5. Kotak Emerging Equity Fund (Regular): 1620.62

Insights:

- Highly concentrated portfolios (HHI >2500) rely on a few sectors and carry sector-specific risk but may offer higher alpha if manager skill is strong
- Well-diversified portfolios (HHI <1500) spread risk but may underperform if concentrated bets pay off
- Most equity funds (79.4%) fall into the moderate concentration range, balancing diversification with manager discretion

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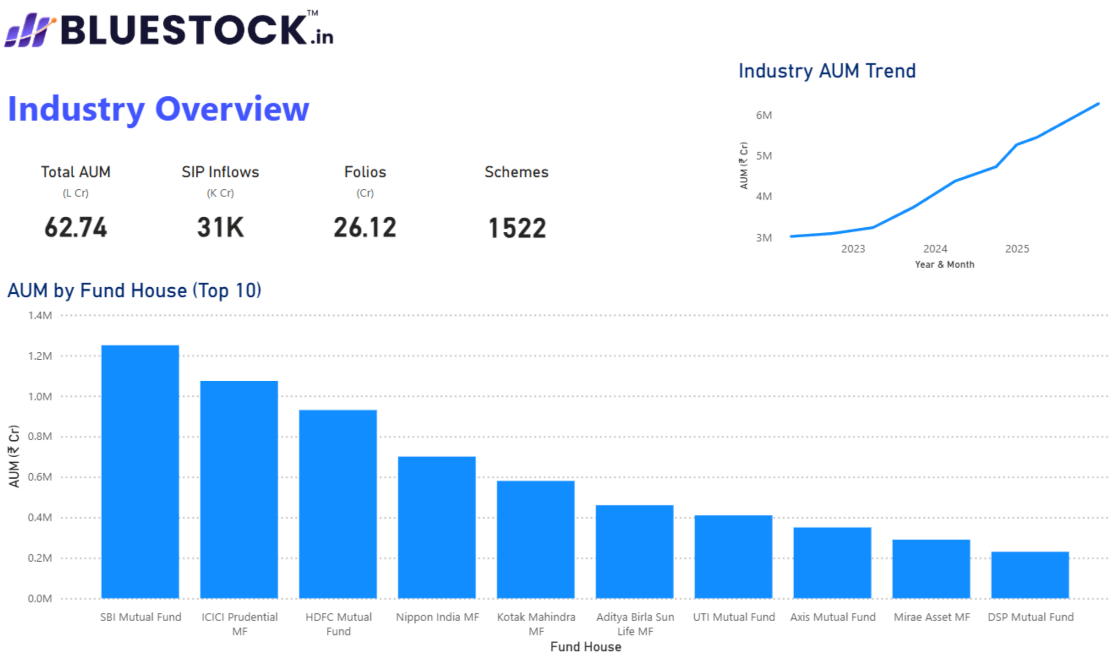
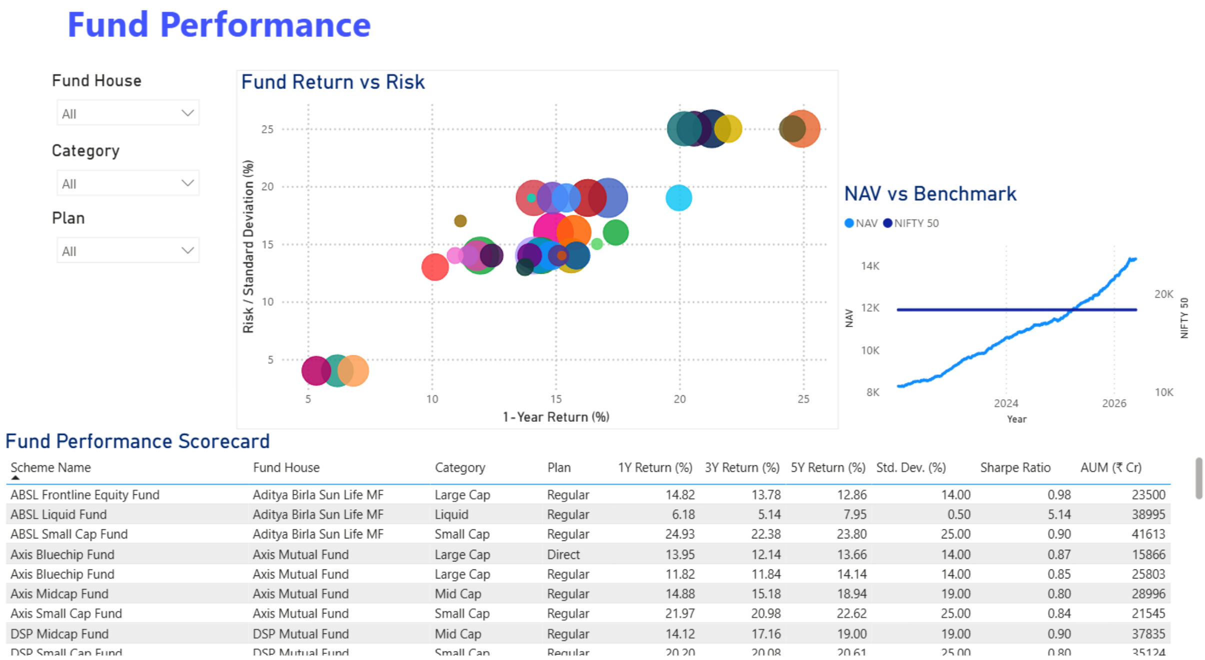
8. DASHBOARD SCREENSHOTS



Industry Overview Dashboard

- AUM growth trends by fund house
- SIP inflows and folio count evolution
- Category-wise inflow distribution
- Geographic and demographic investor profiles



Fund Performance Dashboard

- Fund returns comparison (1Y, 3Y, 5Y CAGR)
- Risk metrics (Sharpe, Sortino, maximum drawdown)

- Fund scorecard rankings
- Benchmark comparison (NIFTY50, NIFTY100)

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9. RECOMMENDATIONS

Based on the analysis findings, the following recommendations are offered:

9.1 For Individual Investors

1. **Risk-Return Alignment**: Investors should select funds aligned with their risk tolerance and investment horizon. The analysis indicates that small-cap funds offer highest absolute returns (21.69% average 3-year) but with highest volatility (25%). Conservative investors may prefer large-cap (12.99% return, 14% volatility) or liquid funds (6.33% return, 0.5% volatility).
2. **Adopt SIP for Discipline**: Monthly SIP participation has surged to 9.35 crore accounts with ₹31,002 Cr in December 2025 inflows. SIP investing promotes discipline and reduces timing risk, particularly for equity-oriented portfolios.
3. **Monitor Risk-Adjusted Returns**: The analysis shows Sharpe and Sortino ratios can identify funds with superior risk-adjusted performance. Investors may consider comparing risk-adjusted returns rather than relying only on absolute returns.
4. **Consider Category Diversification**: Low average correlation (0.0049) among top funds suggests diversification benefits. Combining equity, debt, and liquid funds can improve overall portfolio risk profile.
5. **Evaluate Expense Ratios**: Fund scorecard analysis shows that lower expense ratios correlate with better composite performance. Investors should compare direct vs. regular plan options and indexed fund alternatives where available.

9.2 For Fund Advisors & Asset Managers

1. **Demographic Targeting**: Investors under age 35 represent 56.1% of the investor base. Advisors may develop targeted products and marketing for this demographic. The 56+ segment remains underserved, presenting retirement planning opportunities.
2. **Geographic Expansion**: With 66.3% of investments concentrated in Tier-1 cities, significant growth potential exists in Tier-2 and Tier-3 markets (currently 33.7% of investment).
3. **Gender Outreach**: The 66.5% male, 33.5% female split indicates opportunity for female investor acquisition through targeted financial literacy and product customization.
4. **SIP Continuity**: The low SIP continuity rate (2.2%) among investors with 6+ SIPs suggests opportunity for behavioral interventions (reminders, simplified renewal, incentives) to improve retention.

9.3 For the Analytics Platform

1. **Dashboard Accessibility**: The platform enables comparative fund analysis and performance tracking. Users may leverage the dashboard to compare funds across categories, plans, and risk metrics.

2. **Concentration Risk Monitoring**: The HHI analysis highlights which funds have concentrated sector exposures. Portfolio managers may use this to manage sector-level diversification targets.
3. **Tail Risk Assessment**: VaR/CVaR metrics provide a quantitative basis for downside risk communication and suitability assessment.

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10. LIMITATIONS

The analysis is subject to the following limitations:

1. **Data Scope & Coverage**
   - Analysis covers 40 mutual fund schemes; results may not generalize to the broader fund universe
   - Investor transaction data limited to 2024–2025 period (recent data only); longer historical cohorts would strengthen retention and churn analysis
   - Geographic and demographic data derived from transaction sample; patterns may not reflect entire investor population
2. **Historical Performance & Future Uncertainty**
   - All performance metrics (returns, Sharpe ratios, alpha, beta) are historical and do not guarantee future performance
   - Market conditions (interest rates, volatility, economic growth) may change significantly, affecting fund performance
   - Past outperformance vs. benchmarks does not imply future outperformance
3. **Data Quality & Limitations**
   - Folio and transaction data have inherent limitations in measurement (e.g., partial folios, legacy systems, data reporting delays)
   - NAV forward-fill assumes linear carry-forward across non-trading days; actual pricing or corporate actions may differ
   - Benchmark indices (NIFTY50, NIFTY100) may not be optimal comparisons for all fund categories (e.g., debt, liquid, sector funds)
4. **Analytical Scope**
   - Analysis does not include macroeconomic factors (inflation, interest rates, GDP growth) that drive market returns
   - SIP continuity analysis assumes regular payment patterns; breaks may reflect intentional investor strategy rather than discontinuation intent
   - HHI concentration analysis based on current holdings; portfolio composition changes over time
   - VaR/CVaR assumes historical returns are representative of future risk; extreme events beyond historical data are not captured
5. **Dashboard & Database**
   - Dashboard and database have been designed for the analyzed schemes and timeframes; extension to additional funds or periods may require schema updates
   - Real-time data feeds are not integrated; analysis reflects historical snapshots as of report date

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11. CONCLUSION

The Bluestock Mutual Fund Analytics platform provides a comprehensive, data-driven framework for understanding mutual fund performance, investor behaviour, and industry dynamics. The ETL pipeline successfully processes 10 datasets into a structured database, enabling systematic analysis of 40 schemes over 4+ years.

Key Findings Summary:

1. **Market Growth**: Industry AUM has grown substantially, with top fund houses increasing assets by 98% over three years. SIP participation has reached 9.35 crore accounts, reflecting growing retail investor confidence.
2. **Performance Diversity**: Fund returns span 0.81% to 21.64% annualised, with small-cap and mid-cap funds leading growth. Risk-adjusted returns (Sharpe, Sortino) reveal substantial variation, with liquid and large-cap funds offering efficient risk-return tradeoffs.
3. **Investor Profile**: The typical mutual fund investor is male (66.5%), aged 26–35 (41.1%), based in a Tier-1 city (66.3%), with median SIP amounts of ₹5,000–5,500. Growth potential exists in female investor acquisition and geographic expansion.
4. **Risk Management**: VaR/CVaR analysis reveals that small-cap and mid-cap funds carry substantially higher downside risk than liquid or large-cap funds, supporting informed suitability assessments.
5. **Data-Driven Decisions**: The analytics platform enables comparative fund selection, performance monitoring, and risk assessment through visualizations and scorecard rankings.

The platform serves as a foundation for ongoing monitoring, investor education, and portfolio construction. Regular updates to the database and dashboard will enhance decision-making for investors, advisors, and fund managers alike.